

CHAPTER

47



Immunological and Metabolic Disorders

Learning Objectives

- ⇒ To know about the clinical features of rheumatic fever, rheumatoid arthritis, systemic lupus erythematosus (SLE), ankylosing spondylitis, gout
- ⇒ To know about the relevant physical examinations of the patients of above said diseases
- ⇒ To know about the order and interpretation of laboratory investigations related to above said diseases
- ⇒ To know about the differential diagnosis of arthritis

Introduction

Rheumatic fever is a non suppurative, acute inflammatory complication of group A streptococcal infection, causing combinations of arthritis, carditis, subcutaneous nodules, erythema marginatum and chorea. Chronic rheumatic fever involves all layers of the heart causing major cardiac sequelae referred to as rheumatic heart disease. In spite of its name suggesting an acute arthritis migrating from joint to joint, it is now well known that it is the heart rather than the joints which is first affected. William Boyd years ago, gave the dictum 'rheumatism licks the joint, but bites the whole heart'. Diagnosis is based on Jones criteria to information obtained from history, physical examination and laboratory tests.

- ⊗ A first episode of acute rheumatic fever can occur at any age, but occurs most often between 5 years and 15 years and is usually uncommon below 3 years and above 21 years.
- ⊗ Group A Streptococcal infection (GAS) is the etiological precursor of ARF, but host and environmental factors are important. GAS M proteins share epitopes with proteins found in synovium, heart muscles and heart valve, indicates the probable damage of synovial joints, heart muscle and heart valve. Other factors such as overcrowding, undernutrition, lower socioeconomic status predispose the occurrence of rheumatic fever.
- ⊗ The incidence of the disease is more common in regions such as cold climate, damp weather, places near the rivers and water ways which favours the spread of infection.

Pathology

- ⊗ Initially, the person afflicted with rheumatic fever will have sore throat and upper respiratory tract. After the 2 – 3 weeks of exposure to the microorganism, the person will have cardiac manifestation.
- ⊗ Subsequent episodes of rheumatic fever will worsen the clinical condition of the patient.
- ⊗ Higher incidence of infection is seen after outbreaks of the infection.
- ⊗ There will be increased levels of antibodies to the antigen of Group A streptococcal infection such as antistreptolysin O (ASLO) in the body.

Clinical features

- ⊗ The first episode of acute rheumatic fever generally appears 2 to 3 weeks after streptococcal pharyngitis in the children between 5 to 15 years. With subsequent streptococcal pharyngitis, there is reactivation of the disease and similar clinical manifestations appear with each recurrent attack.
- ⊗ The disease generally presents with migratory polyarthritis and fever.

Consequences

The disease may have the following consequences.

- ⊗ **Carditis** – This is the consequence seen in the person who has developed streptococcal pharyngitis.
- ⊗ **Polyarthritis** – Acute and painful inflammation of the synovial membranes of some of the joints, larger

joints in particular, seen in 90% of the adults. As pain and swelling subside in one joint, other joints tend to get involved, producing the characteristic 'migratory polyarthritis' involving two or more joints at a time.

- ⊗ **Chorea (Sydenham's chorea)**– Chorea minor or St Vitus dance is a delayed manifestation of rheumatic fever as a result of involvement of central nervous system. The condition is characterised by disordered and involuntary jerky movements of the trunk and the extremities accompanied by some degree of emotional instability.
- ⊗ **Erythema marginatum**– This non-pruritic erythematous rash is characteristic of RF. The lesions occur mainly on the trunk and proximal parts of the extremities. The erythematous area develops central clearing and has slightly elevated red margins. The erythema is transient and migratory.
- ⊗ **Subcutaneous nodules**: The subcutaneous nodules occur more often in children than in adults. These nodules are small (0.5 – 2.0 cm in diameter), spherical or ovoid and painless. They are attached to deeper structures like tendons, ligaments, fascia or periosteum and therefore often remain unnoticed by the patients. These are observed in extensor surfaces of the wrists, elbows, knees and ankles.
- ⊗ This is associated with fever, arthralgia, laboratory findings of elevated ESR, raised CRP and leucocytosis.

Complications of rheumatic fever

Death may occur because of cardiac failure, bacterial endocarditis and embolism.

- ⊗ **Cardiac failure**: This is the most common cause of death from rheumatic heart disease. In young patients, cardiac failure occurs due to the chronic valvular deformities, while in older patients, coronary artery disease may be the cause of the cardiac failure.
- ⊗ **Bacterial endocarditis**: The acute and subacute type may supervene due to inadequate use of antibiotics.
- ⊗ **Embolism**: Embolism usually arises from left atrium and its appendages, in association with mitral stenosis. The organs most commonly affected are brain, kidneys, spleen and lungs.
- ⊗ **Sudden death**: This may occur as a result of ball thrombus in the left atrium or due to acute coronary insufficiency in association with aortic stenosis.

Diagnosis of rheumatic fever

Revised Jones criteria for the diagnosis of rheumatic fever (RF):

- ⊗ **Major criteria**
 - Carditis
 - Polyarthritis
 - Chorea (Sydenham's chorea)

- Erythema marginatum
- Subcutaneous nodules

Minor criteria:

- Fever
- Arthralgia
- Previous history of RF
- Laboratory findings of elevated ESR, raised CRP, leucocytosis

Prognosis of rheumatic fever

- ⊗ Prognosis depends mostly on the severity of the initial carditis. Patients with severe carditis during the first episode may have residual heart disease that is often worsened by recurrent episodes of RF.
- ⊗ Murmurs eventually disappear in about half of the patients whose acute episodes were manifested by mild carditis without major cardiac enlargement or decompensation.
- ⊗ Patients who did not have carditis are less likely to have recurrences and unlikely to develop carditis if acute RF recurs.
- ⊗ Sydenham's chorea usually lasts several months and resolves completely in most patients, but about 1/3rd of patients have recurrences. All other manifestations will subside without residual effects.

RHEUMATIC FEVER

DUCKETT-JONES DIAGNOSTIC CRITERIA

MAJOR CRITERIA

"CASES" is the Mnemonic

Carditis
Arthritis
Subcutaneous nodules
Erythema marginatum
Sydenham's chorea

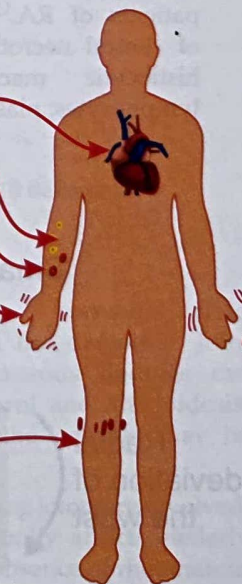
MINOR CRITERIA

"FRAPP" is the Mnemonic

Fever
Raised ESR/CRP
Arthralgia
Prolonged PR Interval
Previous RF

There must be evidence of streptococcal infection plus:
 2 major or 1 major + 2 minor

Figure 208: Duckett Jones Diagnostic criteria



Rheumatoid arthritis (RA) is a chronic systemic autoimmune disease which primarily involves joints. This is characterised by multiple joint involvement, symmetric joint involvement and smaller joint involvement leading to progressive destruction of articular structures associated with systemic symptoms. Diagnosis is based on specific clinical features, laboratory investigations and imaging studies.

Facts of the disease prevalence

- ⊗ Affects 1% of the population
- ⊗ Women are more prone to the diseases 2 to 3 times than men
- ⊗ Onset may be at any age; most often between 35 to 50 years of age group

Etiology: though the disease is autoimmune, the precise cause is unknown, many factors may contribute.

Pathophysiology

- ⊗ Immune complexes produced by synovial lining cells and these are responsible for the immunologic abnormalities in inflamed blood vessels.
- ⊗ Plasma cells produce antibodies (rheumatoid factor) that contribute to these immune complexes.
- ⊗ Macrophages migrate to diseased synovium in early disease and in chronic disease process, lymphocytes infiltrate the synovial tissue.
- ⊗ In chronically affected joints, the thin synovium thickens and develops many villous folds.
- ⊗ Rheumatoid nodules develop in about 30% of the patients of RA. They are granulomas consisting of central necrotic area surrounded by palisaded histiocytic macrophages, all enveloped by lymphocytes, plasma cells, and fibroblasts.

Clinical features

- ⊗ Onset is usually insidious, often beginning with systemic and joint symptoms. Systemic symptoms include early morning stiffness of affected joints, generalised afternoon fatigue and malaise, anorexia, generalised weakness and occasionally low grade fever. The joint symptoms include joint pain, joint swelling, stiffness and deformities.
- ⊗ The disease progresses most rapidly during first 6 years, particularly in the 1st year. 80% of the patients develop some permanent joint deformities within 10 years.
- ⊗ Thus, the course is unpredictable in different diseased individuals.
- ⊗ Joint symptoms are characteristically symmetric. Joint stiffness lasts > 60 minutes after rising in the morning, but may also be seen after prolonged immobility. Involved joints are tender, with erythema, swollen and restricted movements.
- Wrists, index and middle metacarpophalangeal joints are usual afflicted joints
- Other joints involved in the disease are proximal interphalangeal joints, metatarsophalangeal joints, shoulders, elbows, hips, knees and ankle joints.
- Swan neck deformity (hyperextension of the proximal interphalangeal joint and flexion of the distal interphalangeal joint), boutonniere deformity (flexion of the proximal interphalangeal joint and hyperextension of the distal interphalangeal joint), z deformity of the thumb, ulnar deviation of the fingers, Baker's cyst in the posterior aspect of the knee joint are the other frequent sequelae seen during the course of the illness.
- Subcutaneous rheumatoid nodules are seen in 30% of the patients.

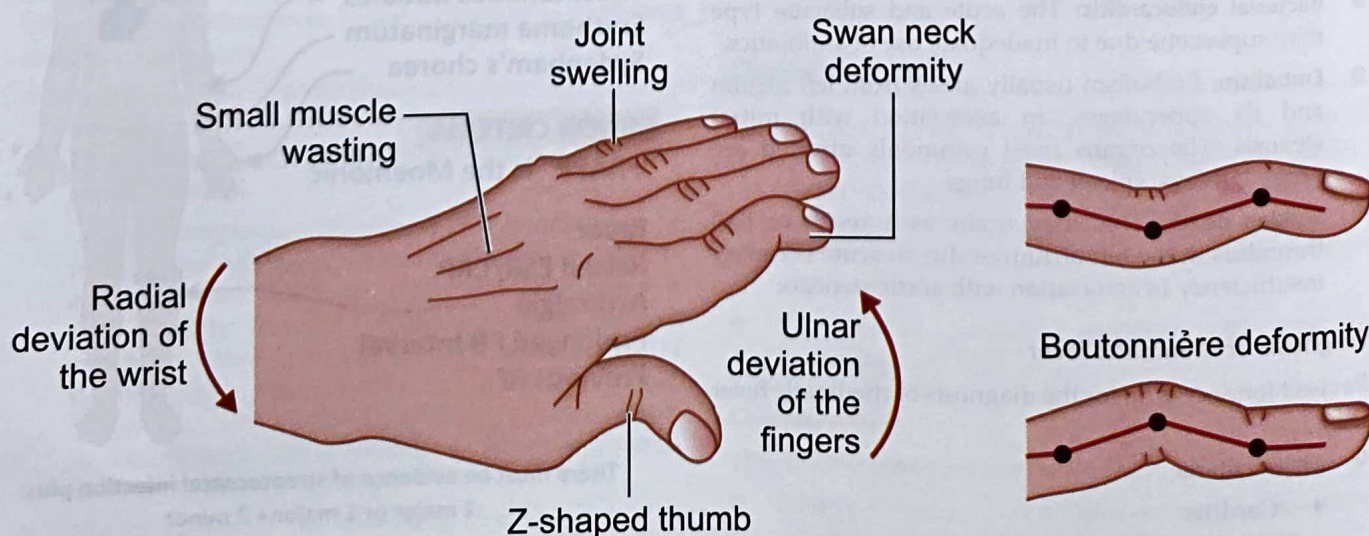


Figure 209: Joints of the patient of rheumatoid arthritis

Image of swan neck and boutonniere deformity: In swan neck deformity, there will be flexion of distal interphalangeal joint and hyperextension of proximal interphalangeal joint. Whereas in boutonniere deformity, there will be flexion of proximal interphalangeal joint and hyperextension of distal interphalangeal joint.

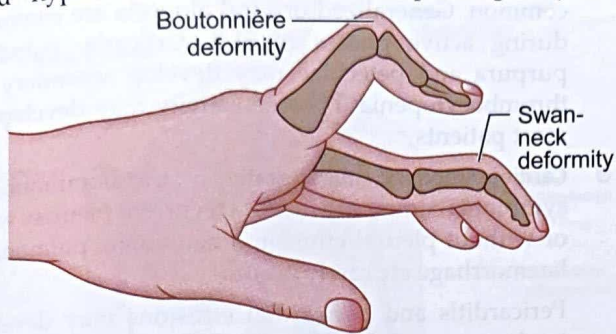


Figure 210: Other images of deformities of rheumatoid arthritis

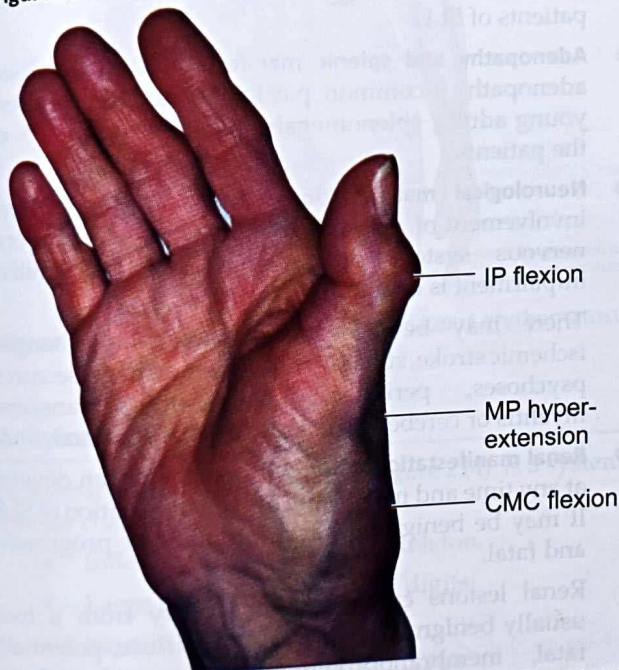


Figure 211: Z deformity in the patient of rheumatoid arthritis:

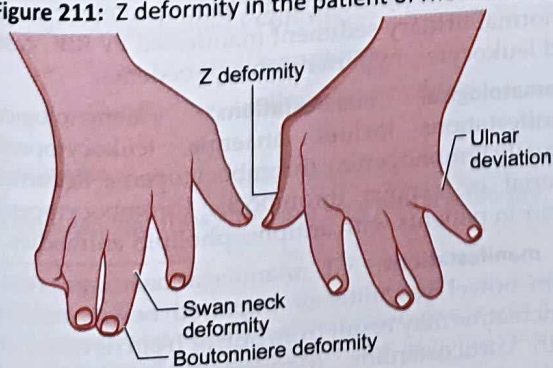


Figure 212: The abnormalities of joints of rheumatoid arthritis
In this image, we can observe ulnar deviation of the fingers and subluxation of metacarpophalangeal joints.

- This is seen in thumb of the patient of rheumatoid arthritis.
- Diagnosis of the disease is by
 - Clinical criteria for the diagnosis of RA:
 - Serum rheumatoid factor (RA test) and anticyclic citrullinated peptide antibody (anti CCP)
 - X rays

2020 ACR-EULAR Classification- Rheumatoid Arthritis

Joint Involvement

1 large joint	0	For patients with at least 1 joint with definite clinical synovitis, not better explained by another disease
2-10 large joints	1	
1-3 small joints, +/- large joints	3	
>10 joints (at least 1 small joint)	5	

Serology (need at least 1)

Rule out:

Negative RF, negative anti CCP Ab	0	Psoriatic arthritis
Low positive RF or low positive anti CCP Ab	2	Viral polyarthritis
High positive RF or high positive anti CCP Ab	3	Gout

Acute Phase reactants (need at least 1)

Normal CRP and normal ESR	0	CPPD
Abnormal CRP or abnormal ESR	1	SLE

Duration of symptoms

< 6 weeks	0	≥ 6/10 definite RA
≥ 6 weeks	1	

Differential diagnosis:

- **Crystal induced arthritis** – synovial fluid examination will confirm the diagnosis. Joint involvement and subcutaneous nodules can occur from gout, cholesterol and amyloidosis. Aspiration or biopsy of the nodules may be necessary.
- **Osteoarthritis** – weight bearing joints are involved, degenerative condition usually afflicts elderly individuals. There will be absence of rheumatoid nodules. There will be involvement of distal and proximal interphalangeal joint in upper limbs. X ray of the affected joint may confirm the diagnosis. In RA, the involvement of distal interphalangeal joint will be absent.

- **SLE** – can be distinguished if there are skin lesions on light exposed areas, hair loss, oral and nasal mucosal lesions, absence of joint erosions in even long standing arthritis. The deformities in this disease are usually reducible as joint erosions are absent during the course of the disease.
- **Sarcoidosis** – though this disease involves the joints, the tissue biopsy is needed for the confirmation of the disease
- **Reactive arthritis** – can be differentiated by antecedent gastro intestinal or genito urinary symptoms; asymmetric involvement of joints, pain in Achilles tendon, sacroiliac joints, large joints are involved.
- **Psoriatic arthritis** – there will be asymmetric involvement of the joints. The differentiation is difficult if there is absence of nail or skin lesions.
- **Ankylosing spondylitis** – may be differentiated by spinal and axial joint involvement; absence of rheumatoid nodules and negative RA factor test.

⊕ **Prognosis:**

- RA decreases life expectancy by 3 to 7 years with heart disease, infection and GI bleeding
- At least 10% of the patients eventually disabled despite vigorous management of the disease

Systemic Lupus Erythematosus

Systemic lupus erythematosus (see Figure 213) (SLE) is a chronic, multisystem, inflammatory disorder of autoimmune aetiology, occurring predominantly in young women. Common manifestations may include arthralgias and arthritis; malar and other skin rashes; pleuritis and pericarditis; renal or CNS involvement; haematologic cytopenias. Diagnosis is done by clinical evaluation and serological evaluation.

- ⊕ 70-90% of the cases are seen in women of child bearing age group. It can affect at any age, including neonates.
- ⊕ Clinical findings may vary. SLE develops abruptly with fever or insidiously over months or years with episodes of arthralgia and malaise.
Vascular headaches, epilepsy or psychoses may be initial findings. Manifestations are referable to any organ system may appear.
- ⊕ **Joint manifestations:** Joint symptoms, ranging from intermittent arthralgias to acute polyarthritis, occur in about 90% of the patients and may precede other manifestations by years.
Most lupus polyarthritis is nondestructive and nondeforming. However, in long standing disease, deformities without bone erosions may develop.
- ⊕ **Skin and mucous membrane manifestations:** Skin lesions

include malar butterfly erythema (flat or raised) that generally spares the nasolabial folds. The absence of papules and pustules helps in distinguishing from rosacea with SLE. Recurrent ulcers on mucous membranes of hard and soft palate, buccal mucosa, mucosa of gums and anterior nasal septum are common. Generalized or focal alopecia are common during active phases of SLE. Urticaria, palpable purpura and petechiae may develop secondary to thrombocytopenia. Photosensitivity may develop in most patients.

- ⊕ **Cardiopulmonary manifestations:** Cardiopulmonary symptoms commonly include recurrent pleurisy with or without pleural effusion. Pneumonia, pulmonary haemorrhage etc rarely occur.

Pericarditis and pericardial effusions may develop in due course of time. Atherosclerosis in the main vessels is one of the complication observed in the patients of SLE.

- ⊕ **Adenopathy and splenic manifestations:** Generalised adenopathy is common, particularly among children, young adults, splenomegaly may develop in 10% of the patients.

- ⊕ **Neurological manifestations:** This can occur from involvement of any part of the central or peripheral nervous system or meninges. Mild cognitive impairment is common.

There may be headaches, personality changes, ischemic stroke, subarachnoid haemorrhage, seizures, psychoses, peripheral neuropathy, transverse myelitis or cerebellar dysfunction are common.

- ⊕ **Renal manifestations:** Renal involvement can develop at any time and may be the only manifestation of SLE. It may be benign and asymptomatic or progressive and fatal.

Renal lesions can range in severity from a focal usually benign, glomerulitis to a diffuse, potentially fatal, membranoproliferative glomerulonephritis. Common manifestations include proteinuria, an abnormal urinary sediment manifested by RBC casts and leukocytes, hypertension and oedema.

- ⊕ **Haematological manifestations:** Haematological manifestations include anaemia, leukocytopenia (usually lymphopenia); thrombocytopenia. Recurrent arterial or venous thrombosis, thrombocytopenia occur in patients with antiphospholipid antibodies.

- ⊕ **GI manifestations:** GI manifestations can result from bowel vasculitis or impaired bowel motility. Pancreatitis may result from SLE or from its treatment with corticosteroids. Manifestations may include abdominal pain from serositis, nausea, vomiting, bowel perforation and pseudo obstruction.

- ⊕ **Diagnosis** is done by clinical criteria; cytopenias, autoantibodies.

Most common symptoms of systemic lupus erythematosus

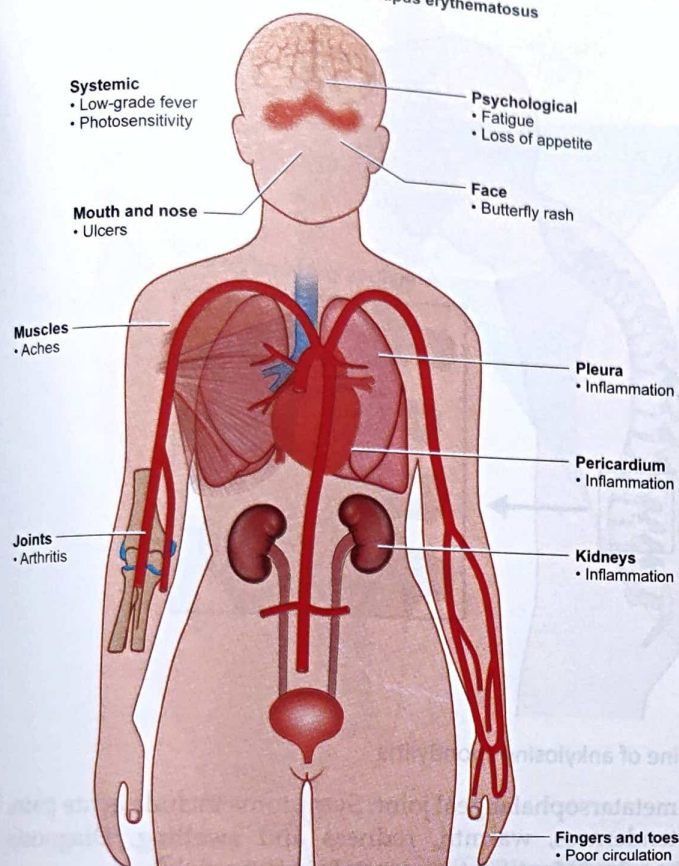


Figure 213: Clinical features of systemic lupus erythematosus (SLE)

Ankylosing Spondylitis

Ankylosing spondylitis (AS) (see Figure 214) is a systemic disease characterized by

- Inflammation of the axial skeleton,
- Large peripheral joints and digits;
- Nocturnal back pain;
- Back stiffness;
- Accentuated kyphosis;
- Constitutional symptoms;
- Cardiac conduction abnormalities.

Diagnosis

Diagnosis requires showing sacroiliitis on X-ray.

- 1 Clinical criteria
 - Low back pain and stiffness which improves with activity for more than three months
 - Limited range of motion of the lumbar spine in both forward and lateral bending.
 - Limitation of chest expansion relative to normal values correlated for age and sex

Radiological criteria

Sacroiliitis grade 2 bilaterally

Sacroiliitis grade 3 to 4 unilaterally

Note: Dugnosis of AS is made if the patient fulfil at least one radiological and one clinical criteria. Source: Copyright © 1994 American Colleges of Rheumatology, an der Linden S. Valkenburg HA, Can A. Evaluation of diagnostic criteria for ankylosing spondyles. A proposal for modification of the New York criteria. Ashes ben 1984.27(4)361-368

- This disease is seen 3 times more in males.

It has no known specific cause.

People who have a gene called HLA-B27 are at increasing risk of developing this disease.

- The most frequent manifestation is back pain, but disease can begin in peripheral joints; especially in children and women.

Other early symptoms are diminished chest expansion from diffuse costovertebral involvement; low grade fever, fatigue, anorexia, weight loss and anaemia.

- Back pain, often nocturnal and of varying intensity, eventually becomes more recurrent.

Morning stiffness, typically relieved by activity and paraspinal muscle spasm may develop.

A flexed, bent over posture eases back pain and paraspinal muscle spasm, thus kyphosis is common in untreated patients.

Severe hip arthritis can develop.

In late stages, the patient has accentuated kyphosis, loss of lumbar lordosis; and flexed bent forward posturing, with compromised pulmonary functions.

- Systemic manifestations may manifest in 1/3rd of the patients.

Recurrent, acute anterior uveitis is common, but self limited.

Neurological signs result from compression radiculitis or sciatica, vertebral fracture or subluxation, or cauda equina syndrome.

Cardiovascular manifestations can include aortic insufficiency, aortitis, angina, pericarditis and conduction disturbances.

Dyspnoea, cough or haemoptysis can result from nontuberculous fibrosis or cavitation of upper lobe of the lung.

- Diagnosis is done by spine imaging (see Figure 215) (bamboo spine).

Blood tests (ESR, C reactive protein, CBC) or modified **New York criteria**¹ may help in the diagnosis.

As far as prognosis is concerned, this is characterized by mild or moderate flares of active inflammation alternating with periods of little or no inflammation.

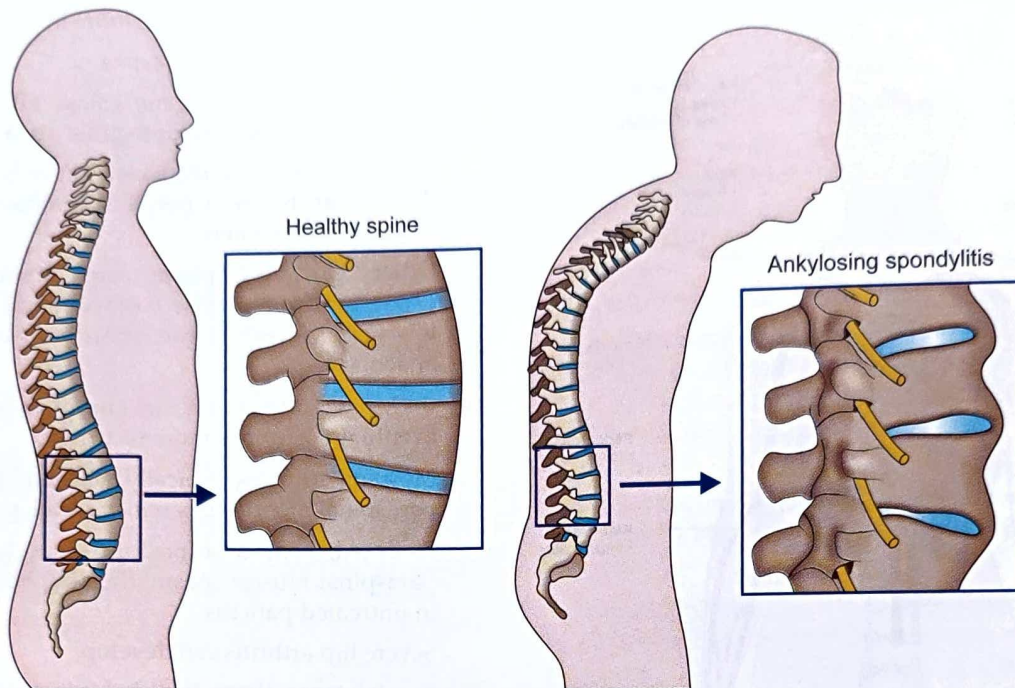


Figure 214: Normal spine and spine of ankylosing spondylitis

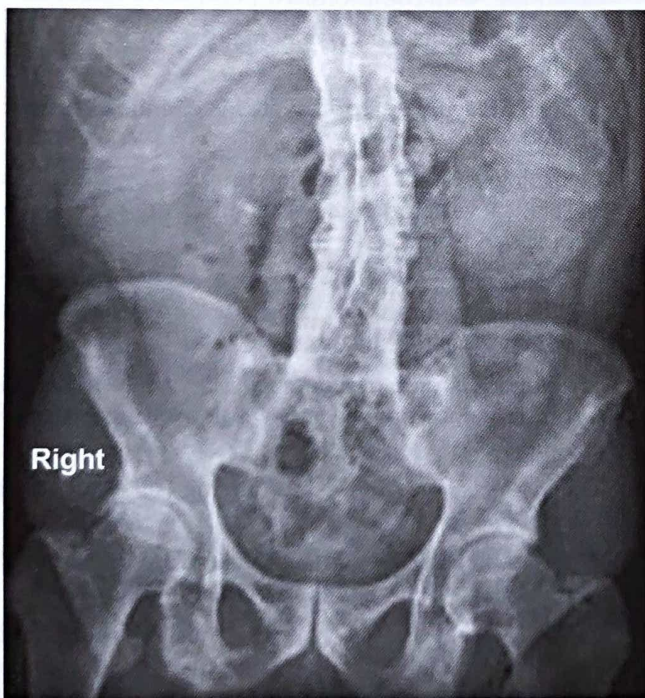


Figure 215: X ray spine of ankylosing spondylitis showing bamboo spine

Gout

Gout (see Figure 216a) is a clinical condition where precipitation of monosodium urate crystals in the tissue can be observed usually in and around the joints. This is causing recurrent acute or chronic arthritis. Acute arthritis is initially monoarticular and often involves the 1st

metatarsophalangeal joint. Symptoms include acute pain, tenderness, warmth, redness and swelling. Diagnosis requires identification crystals in synovial fluid.

Aetiology and pathology

- ⊗ Decreased renal excretion of urate crystals: The most common cause of hyperuricemia in the body. It may be seen in patients with receiving diuretics and in those with diseases that decrease GFR (glomerular filtration rate). Lead poisoning irreversibly damage renal tubules, leading to urate retention.
- ⊗ Increased production of urate crystals, in case of increased nucleoprotein turnover in case of haematological conditions such as lymphoma or leukemia or haemolytic anaemia) and in conditions where increased rates of cellular proliferation and cell death occurs such as psoriasis or cytotoxic cancer therapy or radiation therapy)
- ⊗ Increased intake of purine rich food substances such as alcoholic beverages, sweetened beverages, sea foods, organ meat such as liver etc

Signs and symptoms

- ⊗ Acute gouty arthritis usually begins with sudden onset of pain (often nocturnal). The metatarsophalangeal joint of great toe in most often involved. Rarely, hip, shoulder, sacroiliac sternoclavicular or cervical spine are involved. Pain becomes progressively more severe, usually over a few hours, and is often excruciating. Swelling, warmth, redness and tenderness may be present.

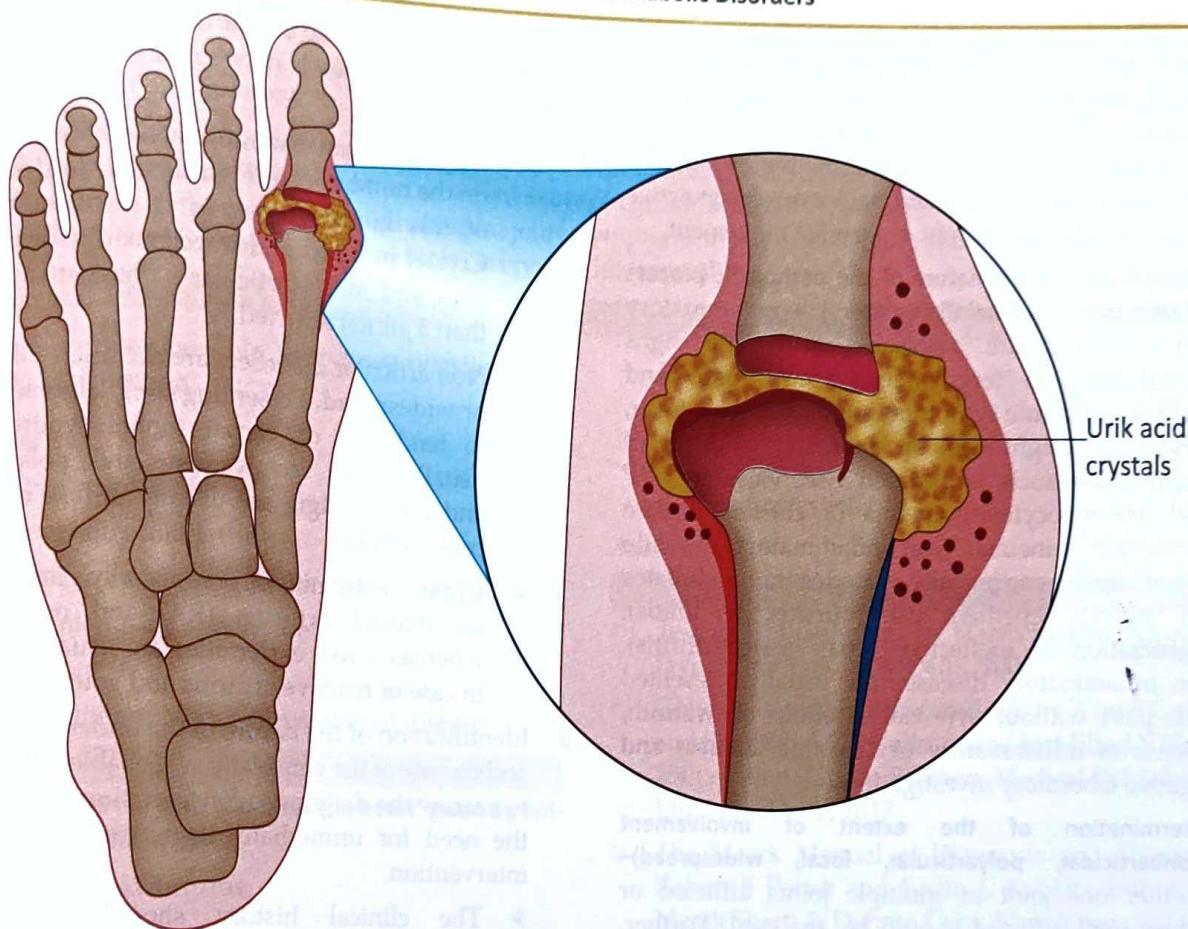


Figure 216a: Gout (Inflammatory Arthritis)

- ⊗ The first few attacks usually affect single joint and last only a few days. Later attacks may affect several joints simultaneously and persist upto 3 weeks if untreated. Subsequent attacks develop after progressively shorter symptom free intervals. Eventually several attacks may occur each year.



Figure 216b: Clinical feature of gout

- ⊗ **Tophi:** (see Figure 216b) Tophi develop most often in patients with chronic gout, but they can never occur

in the patients of acute gout. They are usually firm, yellow or white papules or nodules either single or multiple. They can occur in various locations, commonly the fingers, hands, feet, around olecranon and around Achilles tendon. Tophi may cause deformities.

- ⊗ **Chronic gout:** Pain, deformity and limited joint movement can be observed. Inflammation can be flaring in some joints while subsiding in others. About 20% of the patients may have urolithiasis.
- ⊗ Cardiovascular disease and metabolic syndrome are common in the patients of gout.
- ⊗ Diagnosis is done by clinical criteria and synovial fluid analysis. Serum urate level is being checked to evaluate elevated levels of urate in the blood. As far as prognosis is concerned, the disease is more severe if the early symptoms are observed in the patients in their 2nd and 3rd decade of life. Metabolic syndrome and cardiovascular disease can increase the mortality in patients with gout.

Differential diagnosis of arthritis:

- ⊗ **Anatomic localization of complaint (articular vs. nonarticular)** – first of all, anatomical localisation of the joint should be analysed. Articular involvement should be considered when the person is presented with pain in the joint (joint surfaces) associated with

joint swelling and worsening of the pain on active as well as passive movement of the joint. In case of non articular (ligament, tendon or bursa) involvement, the person will be having pain in the region situated far from the joint surface and pain worsens on active movement and usually not on passive movement.

- ⊗ **Determination of the nature of the pathologic process (inflammatory vs. noninflammatory)** – inflammatory joint swellings are associated with any signs of inflammation (erythema, warmth, pain and swelling), systemic symptoms such as fatigue, fever, rashes and weight loss) or laboratory evidence of inflammation such as elevated ESR, elevated CRP level, thrombocytosis, anemia of chronic disease or hypoalbuminemia. Non inflammatory musculo skeletal diseases are usually related to trauma (rotator cuff injury), repetitive use (bursitis, tendinitis), degeneration or ineffective repair (osteoarthritis). Non inflammatory diseases are usually presented with pain without synovial swelling or warmth, absence of inflammatory or systemic features and negative laboratory investigations.
- ⊗ **Determination of the extent of involvement (monoarticular, polyarticular, focal, widespread)** – whether one joint or multiple joints afflicted or diffuse joint afflicted should be analysed. Further, symmetrical or asymmetrical joint afflicted should be evaluated. For example, when one individual is presented with joint pain with swelling and fever, then probable diagnosis favours septic arthritis. When the person is presented with asymmetrical multiple joint involvement and fever is observed in case of rheumatic fever. When the person is presented with multiple, symmetrical smaller joint involvement confirms the diagnosis of rheumatoid arthritis.
- ⊗ **Determination of chronology (acute vs. chronic)** – the chronology of events is an important diagnostic feature and divided into onset, evolution and duration. The onset of disease is short in septic arthritis and gout and prolonged in osteoarthritis (OA), rheumatoid arthritis (RA) and fibromyalgia. Acute conditions tend to be infectious, crystal induced or reactive. Chronic conditions tend to be non inflammatory or

immunological. The extent of articular involvement is often diagnostic. It may be either articular or non articular.

- Articular disorders are usually classified based on the number of joints afflicted. The description of this kind of analysis is mentioned above. Crystal induced and infectious diseases such as gout and mono articular or oligoarticular (less than 3 joints) afflicted.
- Non articular disorders are classified either focal or widespread. Presenting complaints secondary to tendinitis or carpal tunnel syndrome are usually focal involvement and poliomyelitis and fibromyalgia are usually diffuse in clinical presentation.
- Upper extremities are usually involved in rheumatoid arthritis (RA) and osteoarthritis whereas lower extremities are usually involved in case of reactive arthritis and gout at the onset.
- ⊗ Identification of the nature of the underlying process and the site of the complaint will enable the examiner to narrow the diagnostic considerations and to assess the need for immediate diagnostic or therapeutic intervention.
- The clinical history should identify the precipitating events in the manifestation of the disease. Certain comorbidities usually predispose certain typical or atypical clinical presentation.
- Diabetes mellitus may predispose carpal tunnel syndrome, renal insufficiency may predispose gout, similarly, psoriasis (psoriatic arthritis), myeloma (low back pain), osteoporosis (pathological fracture).
- ⊗ Formulation of a differential diagnosis – a variety of musculo skeletal diseases may be presented with systemic features. For example, fever (SLE or infection), skin rashes (SLE or psoriatic arthritis), nail abnormalities (psoriatic or reactive arthritis), myalgia (fibromyalgia or myopathy), weakness (polymyositis or neuropathy).
- ⊗ Thus, while analysing the case of musculo skeletal disease, elaborate analysis of the patient is necessary.

Key summary of the chapter

- ⊗ Rheumatic fever is a nonsuppurative, acute inflammatory complication of group A streptococcal infection, causing combination of arthritis, carditis, subcutaneous nodules and erythema marginatum. Diagnosis is based on Jones criteria to information gleaned from history, examination and laboratory investigations.
- ⊗ Rheumatoid arthritis (RA) is a chronic systemic autoimmune disease which primarily involves joints. This is characterised by multiple joint involvement, symmetric joint involvement and smaller joint involvement leading to progressive destruction of articular structures associated with systemic symptoms. Diagnosis is based on specific clinical features, laboratory investigations and imaging studies.
- ⊗ Systemic lupus erythematosus (SLE) is a chronic, multisystem, inflammatory disorder of autoimmune aetiology, occurring predominantly in young women. Common manifestations may include arthralgias and arthritis; malar and other skin rashes; pleuritis and pericarditis; renal or CNS involvement haematologic cytopenias. Diagnosis is done by clinical evaluation and serological evaluation.

- Ankylosing spondylitis (AS) is a systemic disease characterized by inflammation of the axial skeleton, large peripheral joints and digits; nocturnal back pain; back stiffness; accentuated kyphosis; constitutional symptoms; cardiac conduction abnormalities. Diagnosis requires showing sacroiliitis on x-ray.
- Gout is a clinical condition where precipitation of monosodium urate crystals in the tissue can be observed usually in and around the joints. This is causing recurrent acute or chronic arthritis. Acute arthritis is initially monoarticular and often involves the 1st metatarsophalangeal joint. Symptoms include acute pain, tenderness, warmth, redness and swelling. Diagnosis requires identification crystals in synovial fluid.

Self-Assessment: Questions for Review

- Long essay questions: No long essay questions are asked in this section.

Short essay questions

- Describe rheumatic fever
- Explain rheumatoid arthritis
- Describe systemic lupus erythematosus (SLE)
- Explain ankylosing spondylitis
- Explain gout
- Describe differential diagnosis of joint disease

- a. Fishes;
- b. Milk products;
- c. Vegetable oil;
- d. Sunflower oil

Answer: (a)

MCQs

1. The term 'licks the joint and bites the heart' given by William Boyd is related to

- a. Rheumatoid arthritis;
- b. Rheumatic fever;
- c. SLE;
- d. Gout

Answer: (b)

2. Swan neck deformity is seen in which of the following disease?

- a. Rheumatoid arthritis;
- b. Rheumatic fever;
- c. SLE;
- d. Gout

Answer: (a)

3. In which part of the body, swan neck deformity is observed in the patients of RA?

- a. Neck;
- b. Knee joint;
- c. Elbow joint;
- d. Fingers

Answer: (d)

4. Appearance of tophus is indicative of which of the following disease?

- a. Rheumatoid arthritis;
- b. Rheumatic fever;
- c. SLE;
- d. Gout

Answer: (d)

5. Which of the following contains excess amount of purine substances?

Chapter resources

- Textbook of pathology by Harsh Mohan, 5th edition, Jaypee brothers, New Delhi, 2005
- Golwalla's medicine for students by Aspi F Golwalla and Sharukh A Golwalla, edited by Milind Y Nadkar, 25th edition, Jaypee Brothers Medical Publishers (P) Ltd, New Delhi, 2017
- The Merck Manual of Diagnosis and Therapy by Robert S Porter and Justin L Kaplan, 19th edition, Merck Sharp & Dohme Corp, New Jersey, 2011
- Chamberlain's symptoms and signs in clinical medicine by Colin Ogilvie and Christopher C Evans, 12th edition, Butterworth-Heinemann Publishers, 1997
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